

Biodiversity

Threats due to natural and anthropogenic activities

In 2014, the estimate was just over seven billion people on the earth. Human population growth is exponential. The more people you have, the more reproduction you have going on. If you have more reproduction happening, then the curve on a graph of population versus time is going to get steeper and steeper to the point where we're looking at about nine billion people by the year 2050. As the population increases, so does the need to exploit the natural resources, these factors that threaten the biodiversity.

There are 7 major anthropogenic activities that needs to be discussed.

1. **Habitat destruction**
2. **Poaching**
3. **Man-wildlife conflicts**
4. **Pollution**
5. **Species introductions**
6. **Global climactic change**
7. **Exploitation of resources**

1. Habitat destruction:

There are various reasons that lead to habitat destruction. They are:

- a. Loss of habitat
- b. Habitat fragmentation
- c. Deforestation
- d. Raw materials
- e. Production of drugs and medicines



Fig: - Habitat destruction

a. Loss of habitat:

Forests and grasslands have been cleared for various reasons such as agriculture, pasturing, human settlement, developmental projects, etc. Because of this kind of activities, we are losing the habitat. Habitat refers to the area where species seek food, get shelter and reproduce. The greatest threat to wild plant and animal species is due to destruction or alteration of their habitat. If an animal's habitat is destroyed or disrupted, it must adapt to the new changes, move elsewhere or die. When it is forced out of its territory, and if it finds a suitable habitat there is a possibility that the habitat is already in use. Consequently, it must compete with the local population of the same species as well as other animals. The other option is that it must migrate into a marginal habitat where it may succumb to predation, starvation or disease. Some organisms such as pigeon, house sparrows, rodents (like rat and mice) and deer flourish in the modified habitats provided by human activities but many others do not. Some habitats are more vulnerable to species extinction, these are called fragile habitats. Coral reefs, oceanic islands and mountain tops are important fragile habitats.

Biodiversity

b. Habitat fragmentation:

What is meant by habitat fragmentation? It is the removal of small sections of the habitat for reasons such as road construction, urbanization, agriculture, resulting in the division of forests into smaller fragments. This kind of activity will affect the biodiversity.

Habitat fragmentation is commonly defined as “the process whereby a large, continuous habitat is both reduced in area and divided into two or more fragments”.

Fragmentation often refers to an extreme reduction in habitat area, but it can also occur when an area is reduced only by a small degree when the original habitat is divided by roads, railroads, power lines, fences, or other barriers obstructing the free movement of species. There are two ways in which the fragments differ from the original habitat – first, fragments have a greater amount of edge for the area of habitat; second, the centre of each habitat fragment is closer to an edge.



Fig: - Habitat fragmentation

In terrestrial and inland water ecosystems, human activities often lead to the fragmentation of habitats.

Habitat fragmentation may speed up the decline of a population and push it to extinction by splitting an existing widespread population into two or more subpopulations, each in a restricted area. The smaller populations often experience various problems associated with their small size like inbreeding depression and genetic drift. Even though a larger area would be able to support a large population, sometimes the smaller fragments of these areas are unable to support smaller groups, which may normally be able to persist for a long period of time. Some animals such as bears and tigers need larger territories, and cannot survive when their habitat is fragmented into smaller sections.

c. Deforestation:

It is a direct cause of extinction and biodiversity. Around 18 million acres of forest are lost every year due to logging and other human practices. Deforestation is happening due to cutting trees for timber, removal of medicinal plants, dam constructions, etc. Deforestation can directly lead to biodiversity loss when animal species that live in the trees no longer have their habitat, cannot relocate, and therefore become extinct. Deforestation can lead certain tree species to permanently disappear, which affects biodiversity of plant species in an environment.

d. Raw materials:

Biodiversity contributes directly or indirectly to many aspects of human well-being, for instance by providing raw materials and contributing to health. Over the past century, many people have benefited from the conversion of natural ecosystems to agricultural land and from

Biodiversity

the exploitation of biodiversity. Wild plants used as raw materials for the production of hybrid seeds as a result of which plant species become endangered.

e. Production of drugs:

Biodiversity plays vital roles in maintaining human and animal health. A wide variety of plants, animals and fungi are used as medicine, essential vitamins, painkillers etc. Natural products have been recognized and used as medicines by ancient cultures all around the world. Many animals are also known to self-medicate using plants and other materials available to them. More than 60% of the world population depend on almost entirely on the plant medicine for primary health care. About 119 pure chemicals are extracted from less than 90 species of higher plants and used as medicines throughout the world, for example, caffeine, methyl salicylate and quinine. Wild plants are used for production of drugs; therefore, several medicinal plants become extinct.

2. Poaching:

Poaching, in law, the illegal shooting, trapping, or taking of game, fish, or plants from private property or from a place where such practices are specially reserved or forbidden. Poaching is a major existential threat to numerous wild organisms worldwide and is an important contributor to biodiversity loss.

The hunting and export of excessive numbers of certain animal species is another important factor leading to dangerous reductions in numbers. There are three main types of hunting:

- i) Commercial hunting – in which the animals are killed for profit from sale of their furs, bones or other parts;
- ii) Subsistence hunting – the killing of animals to provide food for survival; and
- iii) Sport hunting – the killing of animals for recreation. Although subsistence hunting was once a major cause of extinction of some species, it has now declined sharply in most areas. Sport hunting is now closely regulated in most countries; species are endangered only when protective regulation does not exist or are not enforced.

What is the main reason for poaching and killing/hunting of animals?

- Illegal trade of wild-life.
- Despite bans, animals are killed for furs, horns, tusks, skins (crocodile).
- Live specimens are smuggled.
- Existence poaching: Killing animals for food.
- Commercial poaching, hunting & killing animals to sell their products.

Biodiversity

3. Man-wildlife conflicts

Humans have taken care of the living beings which are useful to them through extensive breeding programmes, to derive maximum benefit of their products. During the process, the species have lost certain useful characteristics so much so that these forms cannot survive on their own in nature. A very good example is corn, which is pampered so much by human that if it is left on its own, it cannot survive.

Today human has large herds of domestic animals. These animals can also play a significant part in the reduction of animal populations by overgrazing the land, thus destroying the vegetation on which both they and the wild animals depend. The native wildlife of a particular area is capable of utilising the native plant life much more efficiently than introduced domestic cattle, and is thus much less likely to convert fertile areas into deserts.

The other important parameter is that the domestic cattle are carriers of several diseases which they can transmit to wild animals. For example, the steady rehabilitation of the Great Indian Rhinoceros was seriously hampered by the rinderpest disease which they contracted from the local domestic cattle

They arise when wildlife starts causing immense damage and danger to the man. During such conditions, it becomes very difficult for the forest department to compromise the affected villagers & gain village support for wild life conservations.

E.g.: In, Orissa, Sambalpur village 200 humans killed by elephants. In revenge the villagers killed 100 elephants

How to control this man-wildlife conflicts:

- Tiger conservation projects: Making available tranquilizers guns, binoculars and radiosets, etc., to deal with danger.
- Solar powered fencing instead of electric.
- Cropping near forests should be prevented.
- Sufficient food should be made available for animals within the forest.
- Wild animal hunting rituals should be stopped.

4. Pollution:

All forms of pollution pose a serious threat to biodiversity, but in particular nutrient loading, primarily of nitrogen and phosphorus, which is a major and increasing cause of biodiversity loss and ecosystem dysfunction.

Burning of fossil fuels that releases harmful chemicals, depleting the ozone layer, excessive waste production



Fig: - Man-wildlife conflicts



Fig: - Pollution

Biodiversity

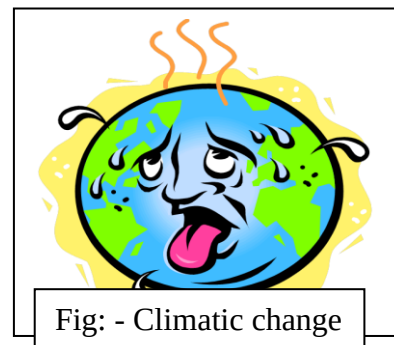
disrupts, fragments, and degrades the ecosystem. Eutrophication, the process of accumulation of nutrients, including nitrogen, in water bodies, often results in water pollution. Nutrient overloads in aquatic ecosystems can cause algal blooms and ultimately a loss of dissolved oxygen, and of life. As ecosystems are impacted, so is the biological diversity.

5. Species introductions:

Introduction of non-native, predatory species that compete for resources can threaten endemic wildlife. Invasive alien species are animals, plants, fungi and microorganisms that entered and established in the environment from outside of their natural habitat. They reproduce rapidly, out-compete native species for food, water and space, and are one of the main causes of global biodiversity loss. For example, Western honey bee, brown rat.

6. Global climactic change:

Climate change caused by global warming represents one of the most serious threats to biodiversity. The high levels of carbon dioxide are likely to cause more extreme weather events like cyclones, hurricanes and droughts. It can also lead to warmer and shorter winters as well as unpredictable monsoons.



The changed atmospheric conditions that result from global warming could create greater numbers of intense storms and prolonged droughts. On the other hand, the expected speed of climate changes coupled with direct loss of natural habitat may prevent some species from adapting quickly enough. They are likely to become extinct, locally or more broadly, and their roles in natural systems will be lost forever.

Rapid artificial climate change does not allow ecosystems and species to adapt. Rising ocean temperatures, diminishing Arctic Sea ice, can affect rising ocean temperatures, affecting marine biodiversity and shift vegetation zones. Rising temperatures are likely to result in widespread ecological change. Many animal and plant species are likely to become extinct as ecosystems adjust to climate change. While adaptable species will survive, the others migrate, the end result will be a lost biodiversity.

7. Exploitation of resources

The unsustainable use of natural resources and overexploitation, which occurs when harvesting exceeds reproduction of wild plant and animal species, continues to be a major threat to biodiversity. Over-hunting, over-fishing, over-harvesting, poaching, and other forms of hunting for profit contribute greatly to loss of biodiversity and death of numerous species.
